IILM UNIVERSITY

B.TECH COMPUTER SCIENCE & ENGINEERING

VIRTUAL JOB SIMULATION REPORT

DELOITTE CYBER JOB SIMULATION

CYBERSECURITY

Web Activity Log Analysis • Network Access • Web Security

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Academic report describing a completed virtual job simulation. This report does not represent employment or a formal internship placement with Deloitte.

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Page references reflect the revised report without a certificate section.

Acknowledgement

I would like to express my gratitude to the learning platform Forage for providing access to the Deloitte Cyber Job Simulation. The simulation offered a structured opportunity to practise cybersecurity concepts through a realistic web-activity investigation scenario.

I am also thankful to IILM University and the faculty members of the Computer Science and Engineering programme for encouraging practical learning alongside academic study. The activities in this simulation helped me connect theoretical concepts such as private IP addressing, authentication, VPN access, and web request analysis with a practical case.

Finally, I acknowledge the creators of the simulation materials and the supporting documentation that helped me understand the scenario and interpret the activity logs. This report is my educational summary of the simulation tasks and learning outcomes.

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Declaration

I, Abhishek Jaiswal, a student of B.Tech Computer Science and Engineering at IILM University, declare that this report titled “Deloitte Cyber Job Simulation — Web Activity Log Analysis” has been prepared by me as an academic summary of the virtual job simulation completed through Forage.

The report describes the activities and concepts covered in the simulation, including review of web activity logs, identification of suspicious request patterns, and interpretation of network access context. Any conclusions are limited to the information provided in the simulation scenario.

I understand that the programme completed was a virtual job simulation and not a formal internship or employment placement with Deloitte. This report is not an official Deloitte document, and I have not represented it as such.

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1. Introduction to the Simulation

Cybersecurity is the practice of protecting systems, networks, applications, and information from unauthorised access, misuse, disruption, or damage. A common source of evidence during a security investigation is a web activity log, which records events such as authentication attempts, requested resources, API calls, timestamps, and client IP addresses.

The Deloitte Cyber Job Simulation, hosted by Forage, introduced a scenario involving Daikibo's manufacturing status dashboard. The task was to review web-request activity and reason about whether a pattern looked consistent with normal user browsing or automated behaviour. The exercise also tested understanding of private networks, intranet-hosted services, and authorised remote access.

The aim of this report is to document the scenario, summarise the investigation steps, present the findings supported by the supplied log patterns, and reflect on the cybersecurity skills practised.

1.1 Objectives

- Understand how web logs support investigation of unusual activity.
- Identify differences between ordinary dashboard use and repeated automated requests.
- Interpret private IP addresses and the purpose of VPN access.
- Recognise a normal sequence of login, page-resource loading, and API requests.
- Communicate findings accurately while distinguishing evidence from assumptions.

2. Cybersecurity and Log Analysis

2.1 Role of web activity logs

Web activity logs provide a time-ordered record of requests made to an application. Depending on the logging configuration, an entry may include a timestamp, source IP address, user or session identifier, requested path, HTTP method, status code, and response outcome. Investigators can use these fields to reconstruct a sequence of activity and identify deviations from expected behaviour.

2.2 Indicators of unusual behaviour

Potential indicators include repeated requests at exact intervals, high request volume, unusual access times, requests made without the page resources normally loaded by a browser, and repeated unauthorised responses. These signals are not conclusive on their own. Automated monitoring, scheduled scripts, misconfigured applications, or legitimate integrations can also create repetitive traffic, so context and corroborating evidence are important.

2.3 Network access concepts

Private IPv4 address ranges, including 192.168.0.0/16, are used inside private networks and are not directly routable as ordinary public internet addresses. An intranet-only application is generally intended for users connected to the organisation’s internal network. A VPN can provide an authenticated, encrypted connection into that private environment for authorised remote users.

3. Scenario and Investigation Objectives

3.1 Scenario overview

The simulation concerned Daikibo’s manufacturing status dashboard, which was described as being hosted on the company intranet. The dashboard displayed machine status information from multiple factory locations. The scenario raised concern about whether unusual access patterns could help explain the exposure of sensitive production-related information.

3.2 Questions addressed

The investigation focused on whether the dashboard was directly accessible from the public internet, which user ID showed the most suspicious activity, what access method was intended for remote employees, how the log was organised, and what request sequence was typical of normal dashboard browsing.

3.3 Expected browsing behaviour

The typical request sequence described in the simulation was: (1) a user logs in, (2) the browser loads dashboard page resources, and (3) the page makes API requests for machine statuses. The scenario stated that the dashboard did not continuously poll for updates or receive push updates; a user had to refresh the page to see the latest status. This context helped distinguish regular hourly API requests from ordinary interactive use.

4. Analysis of Web Activity Logs

4.1 Reviewing the structure

The supplied activity log was organised into blocks associated with unique internal IP addresses. Reviewing each block made it possible to compare timestamps, user IDs, request types, and the order in which endpoints were accessed. This structure is useful for grouping activity by a client address, although one IP address may represent more than one device or user in some network environments.

4.2 Suspicious user pattern

The user ID identified in the simulation as showing the most suspicious activity was **mdB7yD2dp1BFZPontHBQ1Z**. The standout pattern was repeated API requests for machine statuses at consistent hourly intervals, without the usual sequence of dashboard page-resource loading that would normally accompany a person opening the dashboard.

4.3 Why the pattern appeared automated

Human browsing is typically irregular: users open pages, interact with controls, and make requests at varying intervals. The repeated status queries in this scenario occurred at a consistent cadence, suggesting a scheduled or scripted process. The pattern is best described as evidence of possible automation; it does not, by itself, identify who created the process or establish malicious intent.

4.4 Authentication outcomes

The log scenario also included requests that received unauthorised responses after a session was no longer valid. Repeated requests after authentication failure can be relevant because they may indicate that an automated process continued running without handling session expiry correctly. A real investigation would confirm this against authentication, endpoint, and VPN records.

5. Findings and Interpretation

The following conclusions are supported by the scenario and the log patterns reviewed:

• **Dashboard location:** the dashboard was described as an intranet-only service, not a public website.
• **Remote access:** VPN tunnelling was the intended method for authorised users outside the internal network.
• **Source addresses:** recorded requests came from private 192.168.0.x addresses, supporting an internal-network context rather than direct access from a public internet client.
• **Suspicious identity:** the user ID highlighted by the simulation was mdB7yD2dp1BFZPontHBQ1Z.
• **Request cadence:** regular hourly API status checks without normal page-resource loading were the strongest indication of automated use.
• **Evidence limits:** the pattern alone does not prove that an external attacker directly accessed the dashboard or establish the operator's identity.

The best-supported interpretation is that an automated process queried machine-status endpoints from within the internal network context. Additional evidence would be needed to determine whether the activity was authorised, misconfigured, or malicious.

5.1 Evidence versus assumptions

Good security reporting separates observed facts from hypotheses. The timestamps, request sequence, source IP range, and response codes are observable evidence. The identity of the person operating the process, the reason for the activity, and whether data was intentionally disclosed are hypotheses that require independent confirmation.

6. Skills Practised and Learning Outcomes

6.1 Technical skills

• **Log analysis:** reviewing timestamps, source IPs, request paths, user IDs, and response patterns.

• **Network fundamentals:** understanding private IP ranges, intranet services, and VPN-based remote access.

• **Web security:** interpreting authentication behaviour and API requests.

• **Anomaly detection:** recognising highly regular request patterns that differ from normal browsing.

• **Incident reasoning:** evaluating whether a pattern warrants further investigation and identifying what evidence is still missing.

6.2 Professional skills

The exercise also encouraged structured problem-solving, attention to detail, careful interpretation of technical evidence, and concise communication of findings. These skills are useful in security operations, IT support, application troubleshooting, and incident-response workflows.

6.3 Overall learning outcome

The simulation demonstrated that log analysis is not simply counting requests; it requires understanding how the application is expected to behave and comparing that baseline with actual activity. A repeated request pattern becomes more meaningful when combined with context about dashboard refresh behaviour, authentication, and network boundaries.

7. Limitations and Recommendations

7.1 Limitations

This report is based on a virtual learning scenario and the activity log provided for that exercise. It does not constitute a full real-world incident investigation. The log pattern alone cannot confirm whether the process was authorised, which individual controlled it, or whether a breach occurred. Private IP addresses identify a private network context, not a person's physical location.

7.2 Recommended follow-up in a real investigation

• Correlate web requests with identity-provider, VPN, and authentication logs.
• Confirm the device and account associated with the source address at the relevant times.
• Check whether an approved scheduled job or integration explains the hourly cadence.
• Review API permissions and ensure machine-status endpoints require appropriate authentication and authorisation.
• Preserve relevant logs and timestamps before making changes.
• If unauthorised activity is confirmed, follow the organisation's incident-response process and rotate credentials or revoke sessions as appropriate.

7.3 Preventive measures

Useful controls include least-privilege API access, multi-factor authentication, centralised logging, alerting for unusual request cadence, secure session handling, and periodic review of service accounts and scheduled jobs. Monitoring should minimise false positives by accounting for approved automation.

8. Conclusion

The Deloitte Cyber Job Simulation provided practical exposure to a basic web-security investigation. By reviewing the structure of the web activity log, identifying regular hourly machine-status API requests, and interpreting the private IP address context, the exercise illustrated how logs can reveal activity that differs from normal user behaviour.

The key learning was that suspicious activity should be assessed against the expected behaviour of the application. In this scenario, the consistent hourly requests without ordinary page-resource loading supported the conclusion that an automated process was querying the dashboard from an internal network context. The evidence did not independently prove the identity or intent of the operator, nor did it alone establish a confirmed breach.

Overall, the simulation strengthened my understanding of log analysis, networking, VPN access, authentication, and evidence-based cybersecurity reporting. These concepts provide a foundation for further study and practical experience in cybersecurity.

References

1. Forage. *Deloitte Cyber Job Simulation*. Virtual job simulation and completion certificate, October 2026.

2. Daikibo manufacturing status dashboard scenario and web activity log supplied as part of the simulation exercise.

3. Internet Engineering Task Force (IETF). *RFC 1918: Address Allocation for Private Internets.* Background reference for private IPv4 addressing.

4. National Institute of Standards and Technology (NIST). *Computer Security Incident Handling Guide (SP 800-61).* Background reference for evidence-based incident investigation and response.

Submission reminder: attach the original Forage certificate separately if your college requires it. Confirm with your faculty that a virtual job simulation is acceptable for the internship requirement.